# Wei-Chen Chen

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## EXPERIENCE

### *U.S. Food and Drug Administration, USA.*

- 08/2016–present, Mathematical Statistician, Center for Devices and Radiological Health.
- 08/2014–08/2016, ORISE Fellow, Center for Biologics Evaluation and Research.

### *Department of Ecology and Evolutionary Biology, University of Tennessee-Knoxville, USA.*

- 09/2013–07/2014, Postdoctoral Associate.

### *Computer Science and Mathematics Division, Oak Ridge National Laboratory, USA.*

- 08/2011–08/2013, Postdoctoral Research Associate.

### *Division of Biostatistics and Bioinformatics, National Health Research Institutes, Taiwan.*

- 01/2001–08/2005, Research Assistant.

## EDUCATION

- 08/2005–07/2011, Ph.D., Statistics, Iowa State University, USA.
- 09/1998–06/2000, M.S., Statistics, National Central University, Taiwan.
- 09/1994–06/1998, B.S., Mathematics, National Central University, Taiwan.

## PUBLICATIONS

### *Software Released on CRAN*

- 12 pbdR Packages: **kazaam**, **pbdRPC**, **remoter**, **pbdZMQ**, **pbdPROF** (GSoC sponsored (2013, 2014)), **pbdDEMO**, **pbdNCDF4**, **pbdDMAT**, **pbdBASE**, **pbdSLAP**, **pbdMPI**, and **pmclust**.
- 8 R Packages: **float**, **MixfMRI**, **getPass**, **cubfits**, **QZ**, **EMCluster**, **phyclust**, and **MixSim**.

### *Refereed Papers*

[23] Lu, N., Wang, C., **Chen, W.-C.**, Li, H., Song, C., Tiwari, R., Xu, Y., Yue, L.Q. (2021), "Propensity score-integrated power prior approach for augmenting the control arm of a randomized controlled trial by incorporating multiple external data sources," *Journal of Biopharmaceutical Statistics*. (To appear at DOI: 10.1080/10543406.2021.1998098).

[22] Wang, C., Rosner, G.L., Bao, T., Lu, N., **Chen, W.-C.**, Li, H., Tiwari, R., Xu, Y., Yue, L.Q. (2021), "Leveraging real-world evidence for determining performance goals for medical device studies," *Statistics in Medicine*. (DOI: 10.1002/sim.9199).

[21] Li, H., **Chen, W.-C.**, Wang, C., Lu, N., Song, C., Tiwari, R., Xu, Y., and Yue, L. (2021), "Augmenting both arms of a randomized controlled trial using external data: An application of the propensity score-integrated approaches," *Statistics in Biosciences*. (DOI: 10.1007/s12561-021-09315-5).

[20] Lu, N., Wang, C., **Chen, W.-C.**, Li, H., Song, C., Tiwari, R., Xu, Y., and Yue, L. (2021), "Leverage multiple real-world data sources in single-arm medical device clinical studies," *Journal of Biopharmaceutical Statistics*. (DOI: 10.1080/10543406.2021.1897994).

[19] **Chen, W.-C.**, Li, H., Wang, C., Lu, N., Song, C., Tiwari, R., Xu, Y., and Yue, L. (2021), "Evaluation of Diagnostic Tests for Low Prevalence Diseases: A Statistical Approach for Leveraging Real-World Data to Accelerate the Study," *Journal of Biopharmaceutical Statistics*, **31**(3), 375–390.

[18] Li, H., **Chen, W.-C.**, Lu, N., Song, C., Wang, C., Tiwari, R., Xu, Y., and Yue, L. (2021), "Mitigating Study Power Loss Caused by Clinical Trial Disruptions Due to the COVID-19 Pandemic: Leveraging External Data via Propensity Score-Integrated Approaches," *Statistics in Biopharmaceutical Research*. (DOI: 10.1080/19466315.2020.1860813).

[17] Li, H., **Chen, W.-C.**, Lu, N., Wang, C., Tiwari, R., Xu, Y., and Yue, L. (2020), "Novel Statistical Approaches and Applications in Leveraging Real-World Data in Regulatory Clinical Studies," *Health Services and Outcomes Research Methodology*, **20**, 237–246.

[16] **Chen, W.-C.**, Wang, C., Li, H., Lu, N., Tiwari, R., Xu, Y., and Yue, L. (2020), "Propensity Score-Integrated Composite Likelihood Approach for Augmenting the Control Arm of a Randomized Controlled Trial by Incorporating Real-World Data," *Journal of Biopharmaceutical Statistics*, **30**(3), 508–520.

[15] Wang, C., Lu, N., **Chen, W.-C.**, Li, H., Tiwari, R., Xu, Y., and Yue, L. (2020), "Propensity Score-Integrated Composite Likelihood Approach for Incorporating Real-World Evidence in Single-Arm Clinical Studies," *Journal of Biopharmaceutical Statistics*, **30**(3), 495–507.

[14] Wang, C., Li, H., **Chen, W.-C.**, Lu, N., Tiwari, R., Xu, Y., and Yue, L. (2019), "Propensity Score-Integrated Power Prior Approach for Incorporating Real-World Evidence in Single-Arm Clinical Studies," *Journal of Biopharmaceutical Statistics*, **29**(5), 731–748.

[13] Schmidt, D., **Chen, W.-C.**, Matheson, M., and Ostrouchov, G. (2017), "Programming with BIG Data in R: Scaling Analytics from One to Thousands of Nodes," *Big Data Research*, **8**, 1–11.

[12] Li, J.X., **Chen, W.-C.** and Scott, J.A. (2016), "Addressing Prior-data Conflict with Empirical Meta-Analytic-Predictive Priors in Clinical Studies with Historical Information," *Journal of Biopharmaceutical Statistics*, **16**(6), 1056–1066.

[11] Schmidt, D., **Chen, W.-C.**, and Ostrouchov, G. (2016), "Introducing a New Client/Server Framework for Big Data Analytics with the R Language," *Proceedings of the XSEDE16 Conference on Diversity, Big Data, and Science at Scale*, **38**, 1–9.

[10] Gilchrist, M.A., **Chen, W.-C.**, Shah, P., Landerer, C.L., and Zaretzki, R. (2015), "Estimating Gene Expression and Codon-Specific Translational Efficiencies, Mutation Biases, and Selection Coefficients from Genomic Data Alone," *Genome Biology and Evolution*, **7**(6), 1559–1579.

[9] **Chen, W.-C.**, Ostrouchov, G., Pugmire, D., Prabhat, and Wehner, M. (2013), "A Parallel EM Algorithm for Model-Based Clustering Applied to the Exploration of Large Spatio-Temporal Data," *Technometrics*, **55**(4), 513–523.

[8] Melnykov, V., **Chen, W.-C.**, and Maitra, R. (2012), "MixSim: An R Package for Simulating Data to Study Performance of Finite Mixture Modeling and Clustering Algorithms," *Journal of Statistical Software*, **51**(12), 1–25.

[7] Schmidt, D., Ostrouchov, G., **Chen, W.-C.**, and Patel, P. (2012), "Tight Coupling of R and Distributed Linear Algebra for High-Level Programming with Big Data," *High Performance Computing, Networking, Storage and Analysis (SCC), 2012 SC Companion:*, 811–815.

[6] **Chen, W.-C.** and Maitra, R. (2011), "Model-Based Clustering of Regression Time Series Data via APECM – An AECM Algorithm Sung to an Even Faster Beat," *Statistical Analysis and Data Mining*, **4**(6), 567–578. (Student Paper Competition Winner).

[5] Carpenter, S., **Chen, W.-C.**, and Dorman, K.S. (2011), "Rev Variation during Persistent Lentivirus Infection," *Viruses*, **3**, 1–12.

[4] **Chen, W.-C.** and Maiti, T. (2007), "Analysis of cDNA Microarray Experiment with External Controls: A Mixed Model Approach," *Calcutta Statistical Association Bulletin*, **59**, 117–135 (by invitation).

[3] Wen, C.-C., Wu, Y.-J., Huang, Y.-H., **Chen, W.-C.**, Liu, S.-C., Jiang, S.-S., Juang, J.-L., Lin, C.-Y., Fang, W.-T., Hsiung, C.A., and Chang, I.-S. (2006), "A Bayes Regression Approach to Array-CGH Data," *Statistical Applications in Genetics and Molecular Biology*, **5**(1), Article 3.

[2] Fan, T.-H. and **Chen, W.-C.** (2005), "A Bayesian Change Points Analysis of the Seismic Activity in Northeastern Taiwan," *Journal of Statistical Computation and Simulation*, **75**(11), 857–868.

[1] Juang, J.-L., Chen, T.-C., Jiang, S.-S., Hsiung, C.A., **Chen, W.-C.**, Chen, G.-W., Lin, S.-M., Lin, J.-H., Chiu, S.-C., and Lai, Y.-K. (2004), "Coupling Multiplex RT-PCR to a Gene Chip Assay for Sensitive and Semiquantitative Detection of Severe Acute Respiratory Syndrome-Coronavirus," *Laboratory Investigation*, **84**(9), 1085–1091.

### *Paper Released on arXiv.org*

[1] **Chen, W.-C.**, Maitra, R. (2021), "A Practical Model-based Segmentation Approach for Accurate Activation Detection in Single-Subject functional Magnetic Resonance Imaging Studies." (arXiv:2102.03639).

### *Conference Presentations*

[12] **Chen, W.-C.**, Li, H., Lu, N., Song, C., Tiwari, R., Wang, C., Xu, Y., Yue, L.Q. (2021), "Evaluation of Diagnostic Tests for Low Prevalence Diseases with Leveraging Real-World Data," 2021 ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Virtual Workshop.

[11] **Chen, W.-C.**, Li, H., Lu, N., Song, C., Wang, C., Tiwari, R., Xu, Y., Yue, L.Q. (2021), "Evaluation of Diagnostic Tests for Low Prevalence Diseases with Leveraging Real-World Data," Biometrics Section, JSM2021, Virtual Conference.

[10] **Chen, W.-C.** (2019), "A Case Study of Analyzing Recurrent and Terminal Events Using Joint Frailty Model," 12th Annual FDA/AdvaMed Medical Devices and Diagnostics Statistical Issues Conference, Washington, DC, USA. (Joint work with Vandana Mukhi, Heng Li, and Yun-Ling Xu).

[9] **Chen, W.-C.**, Schmidt, D., and Ostrouchov, G. (2018), "The pbdR Project: Distributed Computing with R," Section on Statistical Computing, JSM2018, Vancouver, BC, Canada. (Invited).

[8] **Chen, W.-C.** (2018), "Phyloclustering: A Model-Based Approach for Identifying Microbial Populations," Symposium on Data Science and Statistics, Reston, VA, USA. (Invited).

[7] **Chen, W.-C.** and Lu, N. (2017), "A Bayesian Approach Developed for Single Arm Studies using Performance Goals with Subgroups," Section on Medical Devices and Diagnostics, JSM2017, Baltimore,

MD, USA.

[6] **Chen, W.-C.**, Schmidt, D., and Ostrouchov, G. (2016), "Interactive Terabytes with pbdR," UseR! 2016, Stanford, CA, USA.

[5] **Chen, W.-C.**, Ostrouchov, G., Pugmire, D., Prabhat, and Wehner, M. (2012), "Model-Based Clustering Analysis of Large Climate Simulation Datasets," Section on Statistical Computing, JSM2012, San Diego, CA, USA.

[4] **Chen, W.-C.** and Maitra, R. (2011), "A New AECM Algorithm Tuned to A Faster Beat for Model-Based Clustering of Regression Time Series Data," Section on Statistical Learning and Data Mining, JSM2011, Miami, FL, USA. (Student Paper Competition Winner).

[3] **Chen, W.-C.**, Dorman, K.S., Lunney, J.K., and Rowland, R.R.R. (2011), "Model-Based Functional Clustering for Porcine Reproductive and Respiratory Syndrome Virus Data," SCVMMI, Clarke University, Dubuque, IA, USA.

[2] Melnykov, V., **Chen, W.-C.**, and Maitra, R. (2010), "Model-Based Semisupervised Clustering," Biometrics Section, JSM2010, Vancouver, BC, Canada.

[1] **Chen, W.-C.** and Dorman, K.S. (2009), "Twisted Sisters: Disentangling Selection in Overlapping Reading Frames," Section on Statistical Computing, JSM2009, Washington, DC, USA. (Student Paper Competition Winner).

### *Posters*

[12] Yue, L., **Chen, W.-C.**, Li, H., Lu, N., Song, C., Wang, C., Tiwari, R., and Xu, Y. (2021), "Mitigating Study Power Loss in Clinical Studies Disrupted Due to the COVID-19 Pandemic: Leveraging External Data via Propensity Score-Integrated Approaches," 2021 FDA Science Forum, Silver Spring, MD, USA.

[11] **Chen, W.-C.**, Li, H., Lu, N., Song, C., Tiwari, R., Wang, C., Xu, Y. (2020), and Yue, L., "Novel Statistical Approaches for Leveraging Real-World Data to Support Regulatory Decisions," 2020 FDA Scientific Computing Days, Silver Spring, MD, USA.

[10] Yue, L., **Chen, W.-C.**, Li, H., Lu, N., Tiwari, R., Wang, C., and Xu, Y. (2019), "Leveraging Real-World Data for Regulatory Decision-Making: A Novel Statistical Approach for Augmenting Patient Cohort in the Investigational Clinical Studies," 2019 FDA Science Forum, Silver Spring, MD, USA.

[9] **Chen, W.-C.**, Li, H., Lu, N., Tiwari, R., Wang, C., Xu, Y., and Yue, L. (2019), "Leveraging Real-World Data and Incorporating Real-World Evidence: Harnessing Real-World Factors," 12th Annual FDA/AdvaMed Medical Devices and Diagnostics Statistical Issues Conference, Washington, DC, USA.

[8] **Chen, W.-C.**, Li, J., and Scott, J. (2016), "Subgroup Analyses for Count Data using Bayesian Empirical Meta-Analytical Predictive Priors," 2016 ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Washington, DC, USA.

[7] **Chen, W.-C.**, Li, J., and Scott, J. (2016), "Subgroup Analyses for Count Data using Bayesian Empirical Meta-Analytical Predictive Priors," Biopharmaceutical Section, JSM2016, Chicago, IL, USA.

[6] **Chen, W.-C.** and Maitra, R. (2016), "A Novel Model-based Segmentation Approach for Improved Activation Detection in fMRI Studies," ISMRM 24th Annual Meeting, Singapore.

[5] **Chen, W.-C.**, Li, J., Scott, J., and Mintz, P. (2015), "Bayesian-Commensurate Approach for Safety Assessment in Clinical Studies with Count Outcomes," Biopharmaceutical Section, JSM2015, Seattle, WA, USA.

[4] **Chen, W.-C.**, Zaretzki, R., and Gilchrist, M. (2014), "cubfits: an R package for codon usage bias fits," Society for Molecular Biology and Evolution, SMBE2014, San Juan, Puerto Rico.

[3] **Chen, W.-C.**, Ostrouchov, G., Pugmire, D., Prabhat, and Wehner, M. (2012), "Applying model-based clustering for analysis of Community Atmospheric Model output," Conference on Data Analysis (CoDA), Statistical Sciences Group, Los Alamos National Laboratory, Santa Fe, NM, USA.

[2] Dorman, K.S. and **Chen, W.-C.** (2008), "Overlapping Codon Model for Detecting Selection Effects in Overlapping Reading Frames," Fall Conference on Statistics in Biology 2008, Iowa State University, Ames, Iowa, USA.

[1] **Chen, W.-C.**, Lin, C.-Y., Huang, Y.-S., Chang, I.-S., and, Hsiung, C.A. (2004), "Microarray On Line Analysis System (MOLAS): A web platform for array data management and analysis," Intelligent Systems for Molecular Biology, ISMB2004, Glasgow, Scotland, UK.

## SERVICES

### *FDA Advisory Committee*

- Member of Review Team for General Issues Panel: "Clinical Evaluation of Anti-Hypertensive Devices," Circulatory System Devices Panel Meeting, Dec. 5, 2018. (Executive Summary, Presentation, Panel Questions, and 24 Hour Summary) (Public Media)

### *FDA Office of the Chief Scientist (OCS) Intramural Grant Proposal Review*

- *Office of Women's Health (OWH) Intramural Scientific Research Funding Proposal* (FY2020).

### *Journal Editorship*

- 11/2021–present, Associate Editor, *Pharmaceutical Statistics*.
- 07/2018–present, Associate Editor, *Statistical Analysis and Data Mining: The ASA Data Science Journal*.

### *Journal Manuscript Reviews*

- *Pharmaceutical Statistics* (2021, 2017, 2016 ($\times$2), 2015).
- *Journal of Biopharmaceutical Statistics* (2021 ($\times$4), 2020 ($\times$3), 2019).
- *Statistical Analysis and Data Mining: The ASA Data Science Journal* (2021 ($\times$2), 2020 ($\times$2), 2019 ($\times$3), 2018).
- *Statistics in Biopharmaceutical Research* (2021 ($\times$2), 2015).
- *The R Journal* (2021, 2019 ($\times$2), 2018).
- *Statistics in Medicine* (2019, 2018).
- *Psychological Methods* (2018 ($\times$2)).
- *Journal of Computational and Graphical Statistics* (2018).
- *Methods in Ecology and Evolution* (2015 ($\times$2), 2011).
- *Journal of Statistical Software* (2015, 2012).
- *Technometrics* (2013).

### *Conference Manuscript and Poster Reviews*

- *Biopharmaceutical Section Joint Statistical Meeting Poster Competition* (2016 ($\times$2), 2015 ($\times$4)).
- *IEEE International Conference on Networking, Architecture, and Storage* (2012).

### *Conference Organizing Activities*

- Session Organizer: "From Bayesian Statistics to Machine Learning and Artificial Intelligence: Innovation and Adaptation of Safety Assessment," 2018 ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Washington, DC, USA.
- Session Chair: "Clustering Methods for Big Data Problems," Section on Statistical Learning and Data Science, JSM2017, Baltimore, MD, USA.
- Session Organizer: "Personalized Medicine: How Bayesian Subgroup Analysis Plays Its Role," 2016 ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Washington, DC, USA.
- Session Chair: "Dimension Reduction and Semiparametric Modeling," IMS, JSM2011, Miami Beach, FL, USA.

## TEACHING & MENTORING EXPERIENCE

### *Tutorial and Short Courses*

5. **Chen, W.-C.** (2020), "General Statistical Issues in Designing Superiority and Non-inferiority Randomized Controlled Trial," Introductory Statistics Short Course for OHT2 (Office of Cardiovascular Devices), Lecture 3, FDA/CDRH, Silver Spring, MD, USA.
4. **Chen, W.-C.** (2019), "Pooling Data and Meta-analysis," Statistics for Clinical Trials with Application to Medical Devices, Session 5 Lecture 9, FDA/CDRH, Silver Spring, MD, USA.
3. **Chen, W.-C.** (2017), "Pooling Data and Meta-analysis," Statistics for Clinical Trials with Application to Medical Devices, Session 5 Lecture 8, FDA/CDRH, Silver Spring, MD, USA.
2. Ostrouchov, G. and **Chen, W.-C.** (2017), "Essentials of High Performance and Parallel Statistical Computing with R," Continuing Education Course, JSM2017, Baltimore, MD, USA.
1. **Chen, W.-C.**, Zaretzki, R., and Gilchrist, M. (2014), "Fitting Complex Mixed Models using HPC," NIMBioS Tutorial (Computing in the Cloud: What Every Computational Life Scientist Should Know), University of Tennessee, Knoxville, TN, USA.

### *Mentorship and Student Supervision*

8. Chunxiao Wang, Stat PhD, Oregon State University. 2017 Summer at FDA/CDRH/OSB/DBS.
7. Baosheng He, Industrial Engineering PhD, University of Iowa. 2016 Summer at ORNL/JICS.
6. Yuping Lu, EE&CS PhD, University of Tennessee, Knoxville. 2016 Summer at ORNL.
5. Christian Heckendorf, CS MS, Boston University. 2014 Summer at Google Summer of Code.
4. Gaurav Sehrawat, BS, Jaypee Institute of Information Technology, India. 2013 Summer at Google Summer of Code.
3. Cristian Capdevila, CS BS, University of Tennessee, Knoxville. 2012 Summer at ORNL.

2. John Aidoo, Applied Stat MS, Wright State University. 2010 Summer at Iowa State University.
1. Benjamin Deonovic, BCB&Math BS, Iowa State University. 2010 Summer at Iowa State University.

## HONORS & AWARDS

7. 2020 CDRH/FDA Honor Awards: The CDRH Excellence in Scientific Research Award (External Evidence Methods Research Team) – "For the timely development of statistical methods for the transformation of real-world data into real-world evidence that can support regulatory decision-making." With Heng Li, Nelson Lu, Ram Tiwari, Yunling Xu, Lilly Q. Yue, and Chenguang Wang.
6. 2020 FDA Scientific Achievement Award: Excellence in Analytical Science (RWE Methods Group) – "For extraordinary achievements in the timely development and active promotion of novel statistical methods for leveraging real-world evidence to support regulatory decision-making." With Heng Li, Nelson Lu, Ram Tiwari, Chenguang Wang, Yunling Xu, and Lilly Q. Yue. (Public Media)
5. 2016 Significant Event Award (SEA) at the Oak Ridge National Laboratory for "Harnessing HPC Capability at OLCF with the R Language for Deep Data Science." (Public Media)
4. 2011 Vince Sposito Award for Excellence in Statistical Computing, Department of Statistics, Iowa State University.
3. 2011 ASA Statistical Learning and Data Mining Student Paper Competition Award.
2. 2010 Preparing Future Faculty, Iowa State University.
1. 2009 ASA Statistical Computing/Statistical Graphics Student Paper Competition Award.

## COMPUTER SKILLS

- **Statistical Software**: R, JAGS, WinBUGS/OpenBUGS.
- **Programming Languages**: C, Fortran, Perl.
- **Web Development Tools**: PHP, Html, CGI.
- **Database**: MySQL.
- **Operating Systems**: Linux/Unix, Solaris, MS-Windows.
- **Parallel Computing Libraries**: OpenMPI, MPICH, MS-MPI.
- **Network Related**: Apache, SSL, Samba, NFS, NIS, SSH, ZeroMQ.
- **Others**: TeX/LaTeX, Shell Script, vi, Word, Excel, PowerPoint.

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